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Paper 1: QPRapp: A Web-Based Platform for PK/PD Simulations and Early Feasibility Analysi

Bibliographic Information

- **Title**: QPRapp: A Web-Based Platform for PK/PD Simulations and Early Feasibility Analysis.
- **Authors**: Saroj Dhakal et al.
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Paper Type: METHODOLOGY

Quick Take

QPRapp is a free, browser-based Shiny for Python application that enables interactive PK/PD simulation and early feasibility analysis for small molecules and large-molecule biotherapeutics, including TMDD for mono-, bi-, and tri-specific antibodies. It is validated against SimulX and designed to make model-informed drug development accessible to non-computational scientists in early discovery. The tool is best suited for exploratory simulation and hypothesis generation rather than formal estimation or regulatory-grade analysis.

Executive Summary

QPRapp is a web-based, Shiny for Python–implemented simulation platform that enables interactive PK/PD modeling and early feasibility analysis (EFA) for both small molecules and large-molecule biotherapeutics, including TMDD for mono-, bi-, and tri-specific antibodies. The application provides a dropdown-driven interface supporting 1- or 2-compartment PK structures, four indirect response PD models, first-order absorption, IV/EV dosing, single/repeated dosing, and target-occupancy calculations. It is validated against SimulX across representative scenarios and offers CSV export and plot download for downstream analysis. The tool is positioned as an accessible MIDD enabler for non-computational scientists during early drug discovery, with a modular architecture intended to support future extensions (optimization, sensitivity analysis).

Scientific Context & Motivation

Model-informed drug development (MIDD) improves decision-making and resource efficiency, but its application in early drug discovery is limited by the cost, complexity, and coding requirements of existing tools. Commercial platforms (Phoenix WinNonlin, NONMEM, ADAPT, Lixoft, Applied BioMath Assess, SimBiology) require expensive licenses or specialized expertise, while open-source R Shiny apps (ModVizPop, gPKDviz, e-Campsis) either lack TMDD support or restrict multi-specific molecule scenarios to paid versions. As biotherapeutics—particularly multi-specific antibodies—proliferate, there is a growing need for accessible tools to evaluate dose-exposure relationships, target engagement, and early feasibility (e.g., optimal binding affinity and

half-life for a given dosing interval). QPRapp addresses this gap by providing an open, modular, web-based platform that democratizes PK/PD simulation and EFA for project teams with limited computational expertise.

⚡ Methodological Snapshot

QPRapp is a Shiny for Python–based web application that provides a dropdown-driven graphical interface for PK/PD simulation and early feasibility analysis. The backend associates each user-selected model configuration (molecule type, compartment count, target count, PD model type) with pre-written Python ODE model files, which are solved using the LSODA adaptive stiff/non-stiff integrator. The platform supports small-molecule PK and PK/PD models (1- or 2-compartment, four indirect response PD models) and large-molecule TMDD models (1–3 targets, 1- or 2-compartment), with first-order absorption for extravascular dosing and IV/EV routes. For multi-target complexes, the internalization rate is fixed to the minimum among individual drug-target complexes (rate-determining step). The EFA module performs grid-based exploration of binding-affinity ranges (kon/koff for single-target; KD ranges for multi-target) and outputs heat maps of maximum (single dose) or steady-state trough (repeated dose) target occupancy. Results are visualized via matplotlib and exportable as CSV files, with Cmax and AUClast computed automatically.

Estimator Behavior

As a forward-simulation tool, QPRapp does not perform parameter estimation; its numerical behavior is governed by the LSODA solver (adaptive stiff/non-stiff ODE integrator). The paper reports validation against SimulX across multiple scenarios (small-molecule 1- and 2-compartment PK, PK/PD with indirect response, large-molecule TMDD with 1–3 targets, single and repeated dosing, IV and EV routes), with concordant outputs. No formal numerical error metrics (e.g., tolerance settings, convergence diagnostics, runtime benchmarks) are reported, so the accuracy claims rest on visual/graphical agreement with SimulX rather than quantitative discrepancy analysis. The internalization rate for multi-target complexes is fixed to the minimum among individual complexes (rate-determining step assumption), which is a deliberate simplification that may affect accuracy when target kinetics differ substantially.

Validation Design

The application was validated by comparing QPRapp simulation outputs against SimulX, a reference PK/PD simulation platform, across a comprehensive set of scenarios: (1) small-molecule PK with 2 compartments, single IV and EV dosing, and repeated EV dosing; (2) small-molecule PK/PD with 2 compartments, single and repeated EV dosing; (3) large-molecule PK with 1 target and 1 compartment, single IV and EV dosing, and repeated EV dosing; (4) large-molecule PK with 1 target and 2 compartments, single and repeated EV dosing; (5) large-molecule PK with 2 targets and 1 or 2 compartments, single and repeated EV dosing; and (6) large-molecule PK with 3 targets and 1 or 2 compartments, single and repeated EV dosing. Representative scenarios are shown in Figure 3. The validation is qualitative (graphical concordance) rather than quantitative;

no numerical error metrics or statistical tests are reported. The design covers the main model configurations but does not systematically explore parameter-space extremes or edge cases (e.g., very high/low kon, saturating target concentrations, stiff regimes).

Comparison to Alternatives

QPRapp occupies a distinct niche among PK/PD simulation tools. Compared to commercial platforms (Phoenix WinNonlin, NONMEM, ADAPT, Lixoft, Applied BioMath Assess, MATLAB SimBiology), it offers zero licensing cost, a browser-based GUI, and no coding requirement, but lacks their breadth (e.g., non-compartmental analysis, population estimation, complex dosing regimens). Among open-source R Shiny apps (ModVizPop, gPKDviz, e-Campsis), QPRapp's key differentiators are (i) native support for TMDD with up to three targets (mono-, bi-, tri-specific molecules) and (ii) a dedicated early feasibility analysis (EFA) module with heat-map visualization of target occupancy across affinity ranges. e-Campsis supports TMDD but restricts multi-specific scenarios to a paid version. gPKPDsim and A4S/ATLAS mPBPK are limited to specific model classes (SimBiology-based or PBPK). QPRapp's Python/Shiny stack offers better scalability and ML integration than R Shiny alternatives, though it currently supports fewer model structures (1–2 compartments, IV/EV routes only) than several comparators.

☒ Key Findings

QPRapp successfully implements interactive, equation-based PK/PD simulation and EFA in a browser-based interface requiring no coding or commercial license. Key capabilities include: (1) small-molecule PK and PK/PD simulation with 1- or 2-compartment structures and four indirect response PD models (types I–IV); (2) large-molecule TMDD simulation for mono-, bi-, and tri-specific molecules with target occupancy evaluation in central and peripheral compartments; (3) EFA module generating heat maps of target occupancy across affinity (kon/koff, KD) ranges; (4) support for single and repeated dosing via IV and EV routes with first-order absorption; (5) automated computation of Cmax and AUClast; and (6) CSV export and plot download for downstream analysis. Validation against SimulX across multiple scenarios (small-molecule PK/PK/PD, large-molecule TMDD with 1–3 targets, single/repeated dosing, IV/EV routes) demonstrated concordant outputs, supporting the app's reliability for exploratory use.

Strengths & Limitations

Strengths

- Open-source, web-based, and free, removing cost and licensing barriers to MIDD adoption in discovery.
- Unique combination of TMDD support for up to three targets and a dedicated EFA module with heat-map visualization, not available in other open-source tools.
- Modular architecture (UI and backend) designed for extensibility, with planned optimization and sensitivity-analysis modules.

- Validation against SimulX across a broad set of scenarios (small/large molecules, 1–3 targets, 1–2 compartments, single/repeated dosing, IV/EV routes).
- CSV export and plot download facilitate integration with external workflows and reporting.
- Use of LSODA (adaptive stiff/non-stiff solver) provides robust numerical integration for the ODE systems.
- Shiny for Python leverages Python's ecosystem for scalability and future ML integration.

Limitations (Acknowledged by Authors)

- PD response is driven by total drug concentration in the central compartment; no option for other compartments or unbound concentration (planned for future updates).
- EFA for repeated dosing reports target occupancy at trough concentrations, yielding conservative predictions; enhanced flexibility planned.
- No support for complex dosing scenarios such as infusions and loading doses (planned for future integration).
- Current functionality is optimized for exploratory simulations, hypothesis generation, and early-stage feasibility analysis, not for formal estimation or late-stage regulatory use.

Limitations (Expert Review)

- No explicit mathematical specification of the ODE systems (particularly multi-target TMDD equations and occupancy calculations) is provided in the paper, hindering independent verification.
- Validation against SimulX is described qualitatively (visual concordance); no quantitative error metrics (e.g., relative error, tolerance settings, convergence diagnostics) are reported.
- The fixed-minimum internalization-rate assumption for multi-target complexes is a simplification that may introduce bias when target kinetics differ substantially; no sensitivity analysis is provided.
- No public code repository or versioned release is referenced, limiting reproducibility and community contribution.
- The paper does not report computational performance benchmarks (runtime, memory) or scalability limits for the EFA grid search.
- No formal user study or usability assessment is presented to substantiate the claim of accessibility for non-quantitative scientists.

Generalizability

The tool is generalizable to early-stage drug discovery and development across therapeutic areas, for both small molecules and biotherapeutics. However, its scope is intentionally limited to 1–2 compartment models, IV/EV routes, first-order absorption, and indirect response PD models, which constrains applicability to more complex physiological or dosing scenarios. The EFA module is restricted to large molecules. Validation was performed against a single reference (SimulX), and the generalizability of the numerical accuracy to untested scenarios (e.g., extreme parameter values, stiff systems) is not formally established.

Figures & Tables

- **Figure 1**: Schematic of the QPRapp workflow and typical interface, showing the sidebar navigation and conditional pages.
  - *Significance*: Establishes the user journey and the adaptive dropdown-driven design that is central to the app's accessibility.
- **Figure 2**: Model structure schematics for (a) small molecules (central/peripheral compartments, indirect response PD) and (b) large molecules (TMDD with 1–3 targets).
  - *Significance*: Defines the scope of implemented model architectures, clarifying the modeling assumptions and boundaries.
- **Figure 3**: Validation plots comparing QPRapp outputs against SimulX for representative scenarios (small-molecule PK, large-molecule TMDD).
  - *Significance*: Provides the primary evidence of numerical accuracy and reliability of the simulation engine.
- **Figure 4**: Use case I: one-compartment PK/PD model for a small molecule, single oral dose of 5 mg/kg, showing concentration-time and response-time profiles.
  - *Significance*: Demonstrates the small-molecule PK/PD workflow and indirect response model implementation.
- **Figure 5**: Use case II: two-compartment PK model for a large molecule binding to two targets, single IV dose of 5 mg/kg, showing central/peripheral concentrations and target occupancy.
  - *Significance*: Illustrates the multi-target TMDD capability, a key differentiator of the platform.
- **Figure 6**: Use case III: EFA for a bispecific molecule with two compartments, showing heat maps of target occupancy across combinations of binding affinities.
  - *Significance*: Demonstrates the early feasibility analysis module, which is unique among open-source tools and central to the paper's contribution.

Code & Reproducibility Assessment

The application is publicly accessible via shinyapps.io (https://qp-pmxr.shinyapps.io/qprapp/). The paper does not provide the underlying Python source code repository, model equations in explicit mathematical form, or a formal versioned release, which limits full reproducibility and auditability of the implementation.

Future Directions

The authors identify planned extensions: (i) selection of concentrations in compartments other than central and use of unbound drug concentration for PD driving; (ii) enhanced flexibility in occupancy calculations for repeated dosing (currently conservative trough-based); (iii) support for infusion and loading-dose regimens; and (iv) addition of 'Optimization' and 'Sensitivity Analysis' task modules. Recommended follow-up work includes formal numerical validation with quantitative error metrics, publication of the model equations and source code, and sensitivity analysis of the fixed-minimum internalization-rate assumption for multi-target complexes.

Expert Commentary

QPRapp addresses a genuine and persistent gap: the democratization of quantitative pharmacology for discovery-stage teams. The choice of Shiny for Python over R Shiny is defensible given Python's growing dominance in the modeling ecosystem and the need for scalable, ML-integratable interfaces. However, the paper would benefit from explicit mathematical specification of the implemented ODE systems (particularly the multi-target TMDD equations and the occupancy calculation for repeated dosing), quantitative validation metrics (e.g., relative error vs. SimulX), and a public code repository with versioning. The fixed-minimum internalization-rate assumption for multi-target complexes is a notable simplification that warrants sensitivity testing. As a discovery-stage exploratory tool, QPRapp is fit for purpose; its value proposition will strengthen as optimization and sensitivity-analysis modules are added.

Bottom Line

QPRapp is a free, open-source, Shiny for Python–based web application that makes exploratory PK/PD simulation, TMDD modeling, and early feasibility analysis accessible to non-specialists. It is a practical complement to—not a replacement for—full-featured commercial platforms, and is best suited for hypothesis generation and early-stage design decisions in drug discovery.

□ Figures

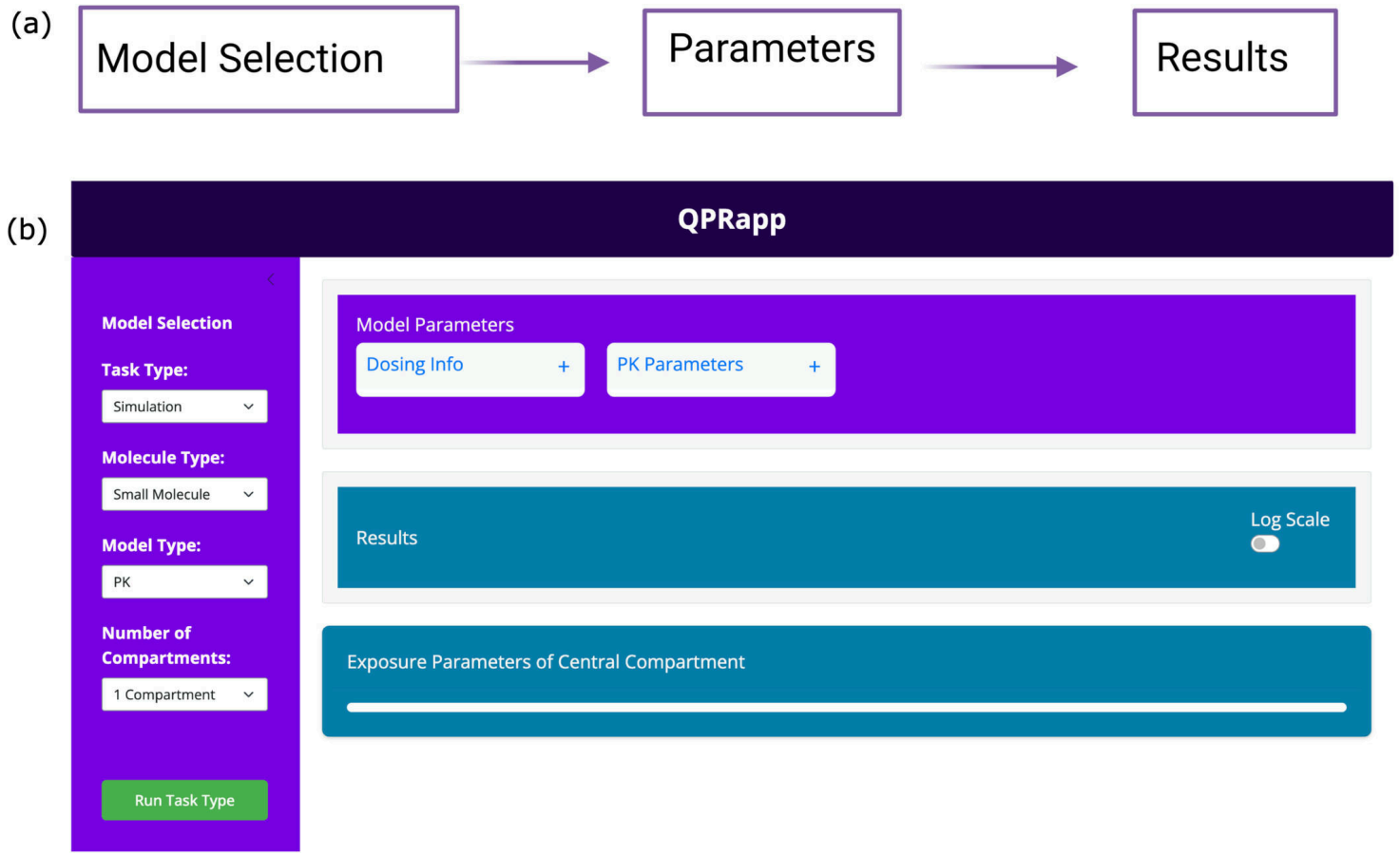


Figure 1: Schematics of the workflow and typical interface of QPRapp. (a) Workflow for the QPRapp. User first selects the appropriate model, inputs the parameters for the

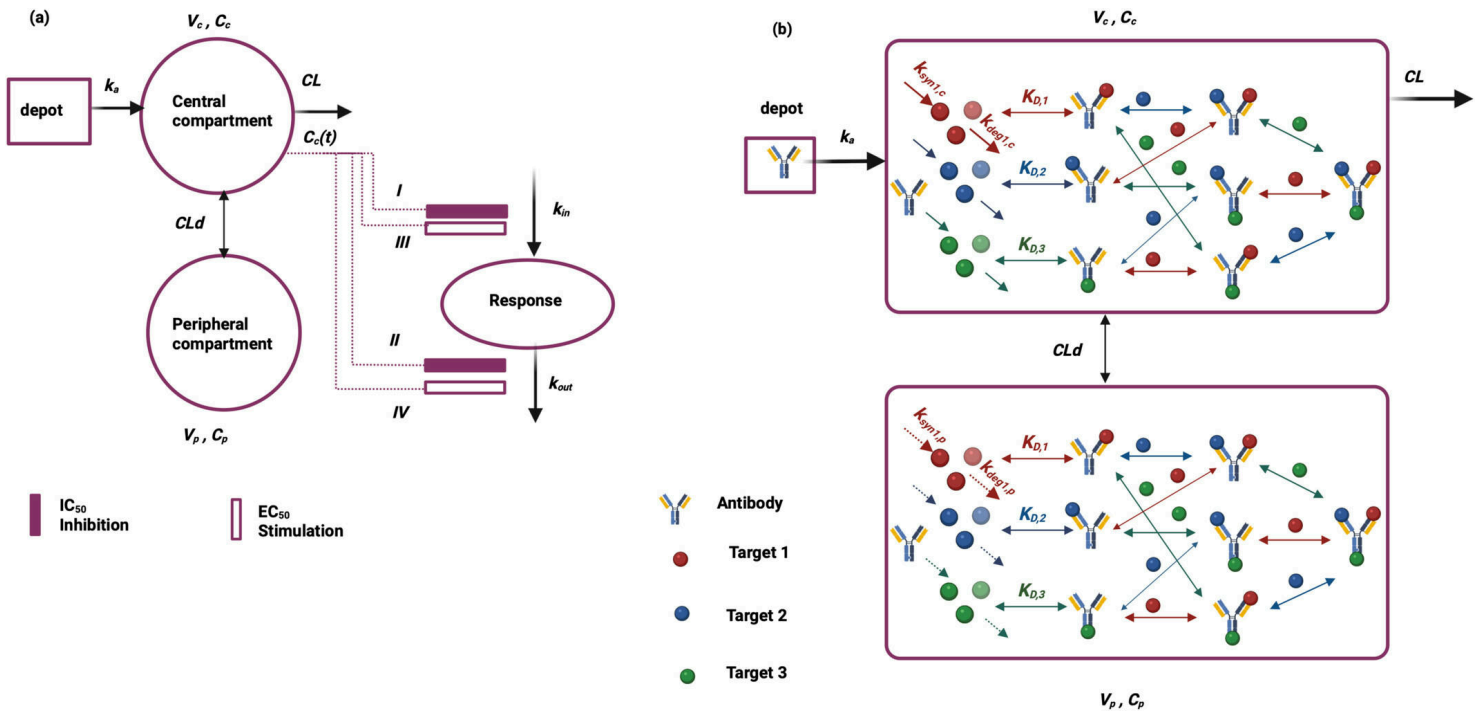


Figure 2: Schematics of model structures in QPRapp. (a) Small molecule model structure with Central and Peripheral compartments. Drug concentration from the central compar

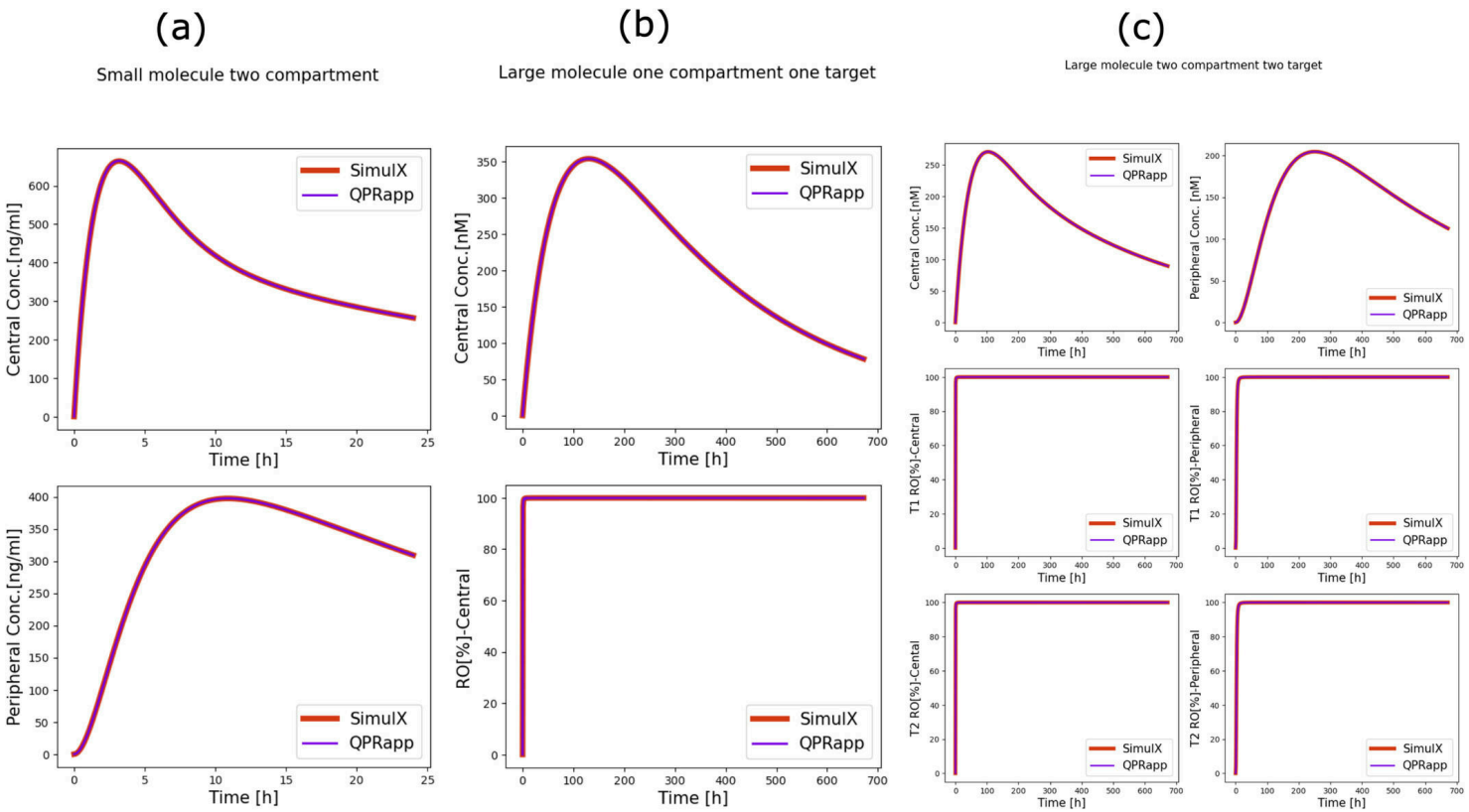
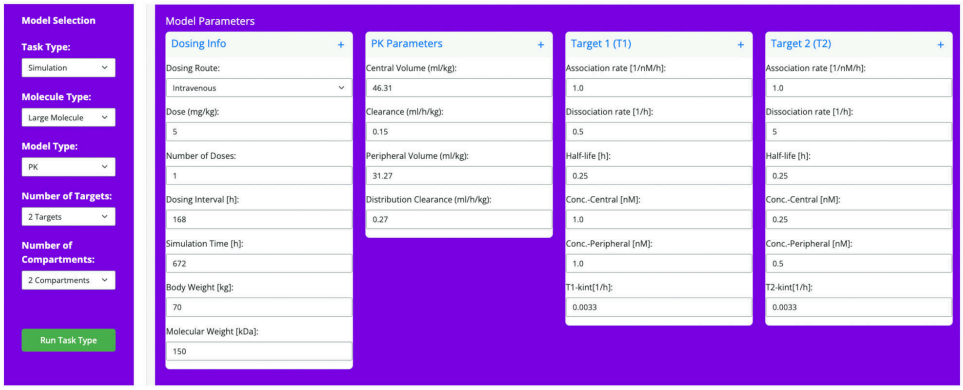


Figure 3: Validation of the QPRapp with SimulX. (a) Small molecule one-compartment model with intravascular dosing in the central compartment. (b) Large molecule two-compa



Figure 4: Small molecule one compartment PK/PD model. (a) Side-bar panel shows the model selected small molecule PK/PD—one compartment. Model Parameters tab shows the para

(a)



(b)

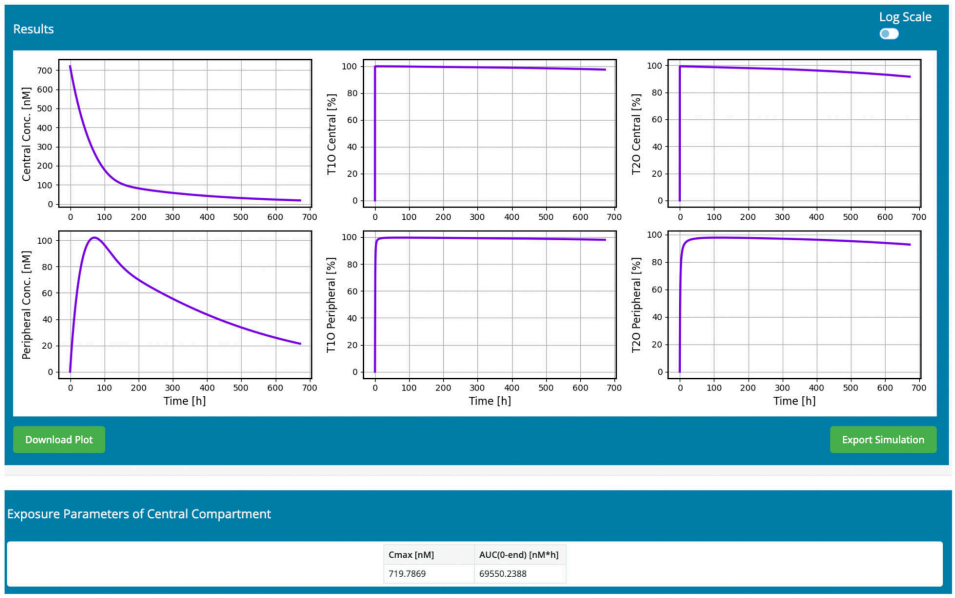


Figure 5: QPRapp interface for large molecule two compartments PK model with two targets. (a) Side-bar panel shows the model selected—Large molecule-PK-2 Targets-2 compart

(a)

(b)

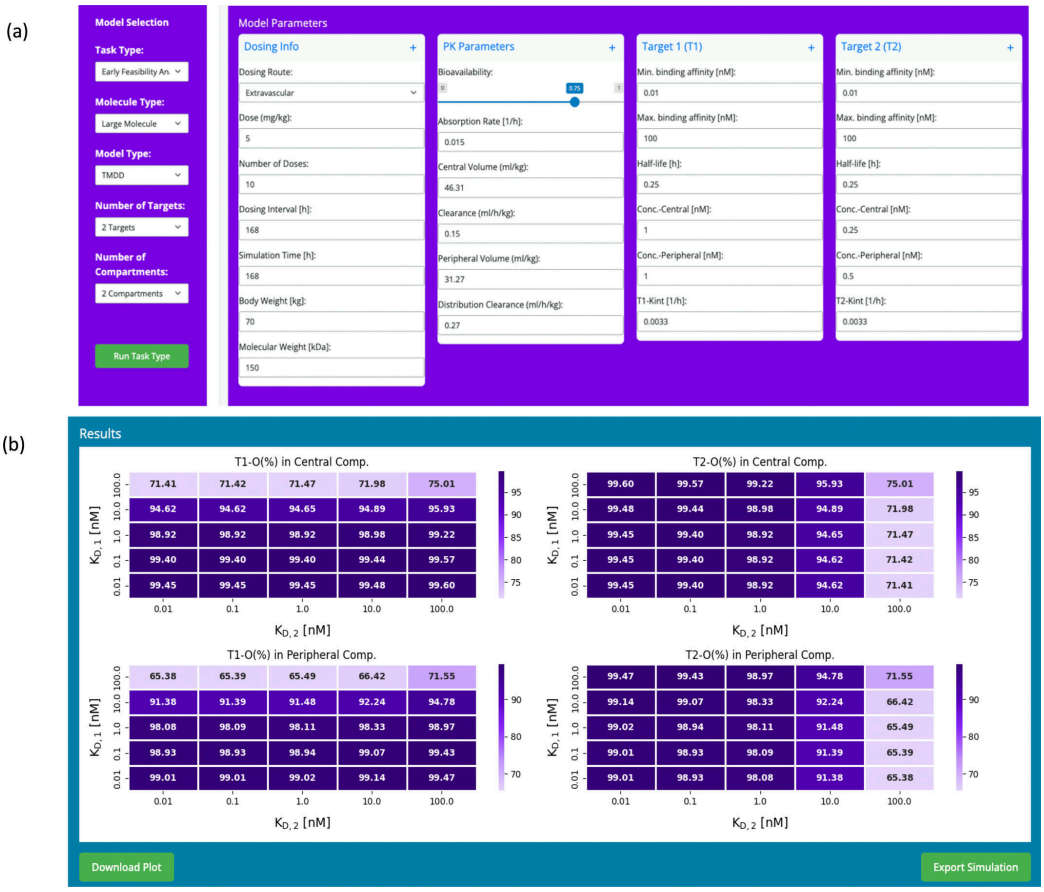


Figure 6: (a) Interface of the QPR app for large molecule/EFA with two compartments and two targets in each compartment. (b) Each heat map shows the target occupancy for d